



$$(W)_{\vec{F}} = \int \vec{F} \cdot d\vec{x} \quad (\times \checkmark)$$

$$v = \frac{dx}{dt}$$

$$ds = v dt$$

$$W = \int \vec{F} \cdot \vec{v} dt$$

JEE
 $m = 2 \text{ kg}$

Q $v = 2t^2$

$t=0 \rightarrow t=2 \quad (W)_F = ?$

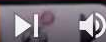
$m_1 \quad \left(\begin{matrix} v=0 \\ t=0 \end{matrix} \right) \Rightarrow \left(\begin{matrix} t=2 \\ v=8 \end{matrix} \right)$

$KE_i = 0 \quad (KE)_f = \frac{1}{2} \times 2 \times 8^2 = 64$

$(W)_{\text{all the force}} = 64$

$$W = \int \vec{F} \cdot \vec{v} dt = \int m a v dt = \int 2 \cdot 4t \cdot 2t^2 dt$$

$$= 16 \times \frac{2^4}{4} = 64$$



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